Tedd Jung

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EDUCATION

Carnegie Mellon University — *Pittsburgh, PA*
Bachelor of Science in Electrical & Computer Engineering — *May 2026*

Honors: Dean's List F23
CS Courses: Computer Graphics (15-362), Computer Systems (18-213), Web Application Development (17-437), Data Structures and Algorithms (15-122), Matrices and Linear Transformations (21-241), Statistical Computing (36-350)
EE Courses: Electronic Devices and Analog Circuits (18-220), Structure and Design of Digital Systems (18-240), Signals and Systems (18-290), Physics II for Engineers (33-142)

WORK EXPERIENCE

Software Engineer Intern — *San Francisco, CA*
Abridge AI Inc. — *June 2025 – Aug. 2025*
- Built a clinician provider search feature using Elasticsearch with full-text, wildcard, and partial matching, eliminating the need for exact queries and boosting search efficiency by 35%, significantly reducing the time searching for providers.
- Designed a full-stack architecture with Firestore integration and automated reindexing through a Google Cloud Function running on a Cloud Scheduler, leveraging TypeScript, NestJS, and Terraform to ensure reliability, maintainability, and scalability.

Teaching Assistant: Web Application Development (17-437/637) — *Pittsburgh, PA*
Carnegie Mellon School of Computer Science — *Aug. 2024 – Present*
- Conducted sprint presentations for a class of 100+ students, guiding teams in developing fully functional full-stack web applications, which were tested and graded as final projects.
- Graded quizzes/assignments, and led lectures, recitations, and office hours on MVC architecture, AJAX, jQuery, JavaScript, React, Django, and REST APIs.

Electrical Integration and Test Engineer Intern — *Pittsburgh, PA*
Astrobotic Technology Inc. — *June 2024 – Aug. 2024*
- Improved overall efficiency of the Avionics team by 20% through designing and testing PCBs, troubleshooting test procedures and writing test scripts for the integrated avionics unit and the power cart unit over the course of two months.
- Developed, tested, and laser welded space manifold heaters over the course of two weeks that will officially be used for the Griffin Lunar Lander, set to launch in Fall 2025.

Research Assistant — *Pittsburgh, PA*
Carnegie Mellon University — *Jan. 2024 – June 2024*
- **Robotics Institute:** Tested and integrated an RTK-GPS system to enhance GNSS accuracy for a lunar rover collecting data on potential human habitation sites on the Moon, contributing to a proof of concept for NASA.
- **MoonRanger:** Collaborated with a team of 60 engineers to develop a lunar rover for NASA's Artemis 3 mission, testing and integrating a NanoSSOC Digital Sun Sensor to enable precise sun position tracking for the communications team.

Firmware Engineer — *Pittsburgh, PA*
Carnegie Mellon Racing — *Feb. 2023 – Present*
- Achieved stateful firmware behavior by diagraming state machines for our firmware architecture, and testing and debugging functions in the Powertrain Thermal Controller using STM32CubeIDE.

PROJECTS

Computer Graphics Software | C++
- Architected a graphics engine with software rasterization, path tracing, and mesh editing capabilities, implementing a GPU-independent rendering pipeline with optimized vertex and fragment processing and utilizing halfedge data structures, Catmull-Clark and Loop Subdivision techniques.

Multiplayer Texas Hold'em Poker Game | HTML/CSS/JS, Django, Redis, AJAX, AWS EC2
- Created a real-time multiplayer poker game, allowing up to 256 concurrent connections at once, by using websockets and deploying it on AWS EC2 with Nginx web server.
- **Github**: https://github.com/tedd1218/chipcity.git

Fantasy Football Bot | Python, Tkinter, BeautifulSoup, pandas, sklearn, kmeans
- Utilized various Python libraries such as BeautifulSoup, Pandas, scikit-learn (including K-means clustering), and other machine learning tools to generate optimized weekly position-based player rankings for Fantasy Football in the NFL.
- **Github**: https://github.com/tedd1218/FantasyFootballBot.git

ADDITIONAL

Languages: Java, Python, C/C++, JavaScript/TypeScript, SystemVerilog/Verilog, MATLAB, SQL/NoSQL, PostgreSQL, R, Swift
Software: VSCode, STM32Cube, AWS, MongoDB, Firestore, Altium, Linux, Elasticsearch
Tools: Git, GitHub, Jira, Confluence, Epsilon3, Linear, LaunchDarkly, Notion, Slack, Docker, Kubernetes, Agile, Codecov
Libraries/Frameworks: React, Next.js, Node.js, Angular, Vue, Django, Tailwind, NumPy, Pandas, scikit-learn, Flask, jQuery